

Pengzhen Li

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Education

Ph.D. in Economics , University of Tennessee, Knoxville	Jan 2020 – May 2027 (expected)
M.A. in Finance , Shandong University, China	Sep 2016 – Jun 2018
B.A. in Economics , Tongji University, Shanghai, China	Sep 2011 – Jun 2015

Research Fields

Trade Policy; Environmental Economics

Publications

- [1] “Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.” With T. Yu et al. *Biomass and Bioenergy*, 208 (2026), 108793.
- [2] “Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.” With T. Yu et al. *Transportation Research Record*, 2678(9) (2024), 639–654.
- [3] “Beef price spread relationship with processing capacity utilization.” With C. Martinez et al. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.
- [4] “Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.” With K. Du and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

Working Papers

- [1] “Trade Restrictiveness Indices with Discriminatory Tariffs.” With James Lake.
- [2] “A Mean-Field Game of Mayoral Endorsements: Theory and Evidence from Brazil and France.” With Matt Van Essen and Luiz Lima.
2024 J. Fred and Wilma Holly Award (best 2nd-year paper).
- [3] “Welfare Analysis of a Forest-Residue Sustainable Aviation Fuel Supply Chain under Endogenous Demand and Price Uncertainty.” With T. Edward Yu.
- [4] “Sustainable Aviation Fuel Production from Forest Residues with Ash Content: A Case Study of Nashville Airport.” With T. Yu et al.

Work in Progress

- “Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value.” With Georg Schaur.

Presentations

- Midwest Economic Theory and International Trade Meetings, Virginia Tech, Blacksburg, VA, Spring 2025.
- Western Regional Science Association (WRSA) Annual Meeting, Hawaii, 2023.
- Western Agricultural Economics Association (WAEA) Annual Meeting, Santa Fe, NM, 2022.
- Southern Agricultural Economics Association (SAEA) Annual Meeting, New Orleans, LA, 2022.

Teaching

Primary Instructor

ECON 351 – Monetary Economics	Spring 2025, 2026
ECON 322 – The Global Economy: Trade and Development	Summer 2025

Teaching Assistant

ECON 511 – Microeconomic Theory	Fall 2023–2025
ECON 581 – Mathematical Methods in Economics	Summer 2024
ECON 512 – Advanced Microeconomics	Spring 2024
AREC 525 – Agribusiness Operations Research Methods	Spring 2022–2023

Research Experience

Graduate Research Assistant, University of Tennessee

Jan 2020 – Aug 2023

- NIFA (USDA) and FAA Office of Environment and Energy (ASCENT) funded projects on forest-residue-based biofuel and sustainable aviation fuel supply chains in the Southeast U.S.
- Designed cost-minimizing and profit-maximizing supply-chain networks using mixed-integer programming (GAMS).
- Conducted facility-siting and investment analyses combining econometric models with the IMPLAN regional impact platform.
- Performed sensitivity and scenario analysis of feedstock quality and market conditions using extended mathematical programming (EMP) and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

Honors and Awards

- J. Fred and Wilma Holly Award (best 2nd-year paper), University of Tennessee, 2024.

Technical Skills

STATA, R, Python, GAMS, MATLAB, SQL, ArcGIS; CFA Level II.

References

James Lake

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Georg Schaur

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Matthew Van Essen

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